Polynomial Division

Exam code: 9MA0

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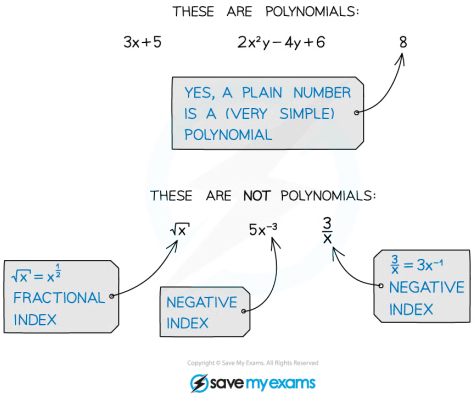
Help me understand

Polynomial division

What is a polynomial?

- A **polynomial** is an algebraic expression consisting of a finite number of terms, with non-negative integer indices only



What is polynomial division?

- Polynomial division is a method for splitting polynomials into factor pairs (with or without an accompanying remainder term)

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

$$x^3 + x^2 - x - 1 = (x+1)(x^2 - 1)$$

$$x^3 + 2x^2 + 3x + 4 = (x-1)(x^2 + 3x + 6) + 10$$

FACTOR PAIR WITH REMAINDER

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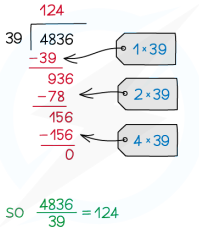
- At A level you will most frequently use it to factorise polynomials, or when dealing with improper (ie 'top-heavy') algebraic fractions

How do I divide polynomials?

- The method used for polynomial division is just like the long division method (sometimes called 'bus stop division') used to divide regular numbers:



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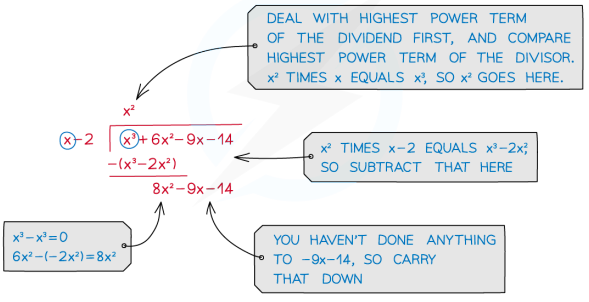
- At A level you will normally be dividing a polynomial dividend of degree 3 or 4 by a divisor in the form **(x ± p)**
- The answer to a polynomial division question is built up term by term, working downwards in powers of the variable (usually ***x***)

- Start by dividing by the highest power term
- Write out this multiplied by the divisor and subtract

> **?** Divide the polynomial $f(x) = x^3 + 6x^2 - 9x - 14$ by $(x - 2)$.



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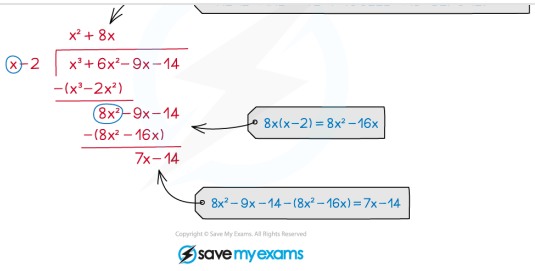
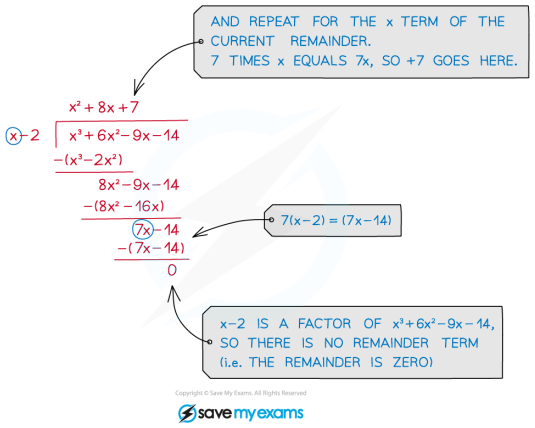
- Continue to divide by each reducing power term and subtracting your answer each time



- Continue until you are left with zero



- If the divisor is not a factor of the polynomial then there will be a remainder term left at the end of the division

Worked Example

Teacher only

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